

Dihedral

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Chapter 1

Introduction

This blueprint formalizes a combinatorial description of curve neighborhoods for the affine Weyl group of type $\tilde{A}_1$, identified with the infinite dihedral group D_∞ .

The strategy is:

1. define chains and degree constraints,
2. characterize the set $A_d(u)$ and its maximal elements,
3. identify curve neighborhoods with translates of those maxima.

All key results are formalized in Lean in `Dihedral/alternatingword.lean`, `Dihedral/Degree'.lean`, and `Dihedral/Neib'.lean`.

Chapter 2

Algebraic Preliminaries

Theorem 1. *The group D_∞ is generated by the two simple reflections $sr(0)$ and $sr(1)$.*

Definition 2. Let $\ell : D_\infty \rightarrow \mathbb{N}$ be the Coxeter length.

Theorem 3. *For every $k \in \mathbb{Z}$,*

$$\ell(sr\,k) = |2k - 1|.$$

Theorem 4. *For every $g \in D_\infty$,*

$$\ell(g) = \text{length}(\text{reducedWord}(g)).$$

Chapter 3

Degrees, Chains, and Order

Definition 5. A degree is a pair $(a, b) \in \mathbb{N}^2$ with product order.

Definition 6. A root is a pair (a, b) such that $a = b + 1$ or $b = a + 1$.

Definition 7. $\text{HasChain } u \ v \ d$ means that v is reached from u by a finite chain of edge moves whose total degree is d .

Theorem 8. If $\ell(u) \leq \ell(v)$, then

$$\ell(uv) = \ell(u) + \ell(v) \quad \text{or} \quad \ell(uv) = \ell(v) - \ell(u).$$

Theorem 9. For vertices u, v ,

$$u < v \iff \ell(u) < \ell(v).$$

Chapter 4

The Set $A_d(u)$

Definition 10. For $u \in D_\infty$ and degree bound d ,

$$A_d(u) := \{v \mid \ell(uv) = \ell(u) + \ell(v), \varphi(v) \leq d\}.$$

In Lean this is `Ad u d`.

Definition 11. `IsMaximalIn v S` means $v \in S$ and every element of S above v is equal to v .

Theorem 12. *If $u \neq 1$, then $A_d(u)$ has a unique maximal element.*

Theorem 13. *If $d.a \neq d.b$, then $A_d(1)$ has a unique maximal element.*

Theorem 14. *For diagonal degree (a, a) , maximal elements of $A_{(a,a)}(1)$ are exactly*

$$r(a), r(-a).$$

Chapter 5

Curve Neighborhood at the Identity

Definition 15.

$$\text{ReachableSet}(u, d) := \{v \mid \exists d' \leq d, \text{HasChain } u \ v \ d'\}.$$

Definition 16.

$$\text{CurveNeighborhood}(u, d) := \{v \mid \text{IsMaximalIn } v \ (\text{ReachableSet}(u, d))\}.$$

Theorem 17.

$$\text{CurveNeighborhood}(1, d) = \{v \mid \text{IsMaximalIn } v \ (A_d(1))\}.$$

Theorem 18. *If $d.a = d.b$, then*

$$\text{CurveNeighborhood}(1, d) = \{s0s1_pow \ d.a, \ s1s0_pow \ d.a\}.$$

Theorem 19. *If $d.a \neq d.b$, then*

$$\text{CurveNeighborhood}(1, d) = \{s_alpha_d \ d\}.$$

Theorem 20. *Complete classification:*

$$\text{CurveNeighborhood}(1, d) = \begin{cases} \{s0s1_pow \ d.a, \ s1s0_pow \ d.a\}, & d.a = d.b, \\ \{s_alpha_d \ d\}, & d.a \neq d.b. \end{cases}$$

Chapter 6

General Base Point and Main Result

Lemma 21. *Under the technical hypothesis `ends_in_s0` u , one gets explicit maximal elements of $A_d(u)$ by case split on `d.a` and `d.b`.*

Theorem 22. *If $v \in \text{CurveNeighborhood}(u, d)$, then*

$$\varphi(u^{-1}v) \leq d.$$

Theorem 23. *If $v \in \text{CurveNeighborhood}(u, d)$, then*

$$u^{-1}v \in A_d(u).$$

Theorem 24. *For every $u \in D_\infty$ and every degree bound d ,*

$$\text{CurveNeighborhood}(u, d) = \{v \mid \exists w, \text{ IsMaximalIn } w (A_d(u)) \wedge v = u \cdot w\}.$$

Proof. The forward direction uses [23](#) to put $u^{-1}v$ in $A_d(u)$, then maximality arguments in the length order. The reverse direction lifts maximal elements of $A_d(u)$ to reachable maximal points via left multiplication. $\square$

Hence $\ell(u \cdot w) = \ell(u) + \ell(w)$, so $w \in A_d(u)$.